Without the cross-modal block: can the bias (c, d) be recovered, and what the noise calibration really measures

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Summary

1. c **cannot be recovered from the main experiment.** This holds whatever is known about σ_A. Recovered and true values correlate at $r \leq 0.38$ (often near 0). The error (RMSE 0.40–0.67) is three to four times the spread of the true values (SD 0.14), and 10–37% of fits end at a bound. Under fusion, c cancels exactly; under CI and switching it acts only through the perceived discrepancy. *Fix $c = 0$ in the bimodal fits.*

2. d **can be recovered only for fusion observers, and only when σ_A and σ_V are known very precisely.**

   - With a log error of 0.05 on both: $r = 0.83$, RMSE 0.19 (true SD 0.27).
   - With a log error of 0.15: $r = 0.65$, RMSE 0.35, so already worse than guessing the mean.
   - For CI and switching observers, never ($r \leq 0.43$, RMSE ≥ 0.44): d trades off against p_c and p_{switch}.

   A per-participant d needs the cross-modal block.

3. **Free bias terms make model selection worse,** even though every simulated participant had a bias. With σ_A known (log error 0.15), the right model is found for 92% without bias terms, 86% with d, and 80% with c and d (fusion only 57%). *Compare models with the no-bias fits.*

4. **Ignoring d costs little in p_c.** The p_c error grows by about 0.2 per unit of d: participants with $d < 0$ get p_c about 0.11 too low, and d between 0 and 0.6 is nearly unbiased. The overall RMSE is the same with and without d in the fit (0.155 vs 0.158).

5. **The big gain is elsewhere: tie σ_A to the unimodal auditory block,** as your joint fit does. Model recovery rises from 83% to 92% (switching 77% → 97%, CI 88% → 93%, fusion 83% → 86%), and the p_c RMSE falls from 0.167 to 0.155 (0.127 at log error 0.05).

6. **The catch: the unimodal block's σ_A is a mixture of two orders** (Section 3). In loud noise most participants are near chance with the test first and sharp with it second. Noise barely changes discrimination with the test second (P105: σ 0.12 at noise 0.5, 0.22 at 2.0). A single σ_A per noise level averages two very different things, and the noise calibration overshot for P103 and P105 because of it. It is now capped at 1.2.

Suggested analysis plan.

- **Main fit:** the joint fit with σ_A and σ_V shared with the unimodal blocks, no bias terms, models compared by AIC.
- **Sensitivity check:** refit with d free, or fixed at a group value, and confirm that model choice and p_c stay the same.

- **If c or d per participant matter scientifically:** only the cross-modal block measures them. With 3 standards, d is measured with an SE of 0.08–0.18 (28 Oct report). It is still in the launcher as an option. A group-level (hierarchical) d would be the alternative.
- **Before relying on shared σ_A:** decide how to handle presentation order (a σ_A per order, or a model of the first interval), and check whether the bimodal data show the same order dependence.

1 What was simulated

`bias_recovery.py`; analysis and figures in `bias_analyse.py`.

- **Participants.** 346 simulated participants: 116 causal inference (CI), 115 forced fusion, 115 switching. Model parameters are drawn as in the 28 Oct report: $\lambda \sim U(0.001, 0.1)$, $\sigma_{A,\text{low}} \sim U(0.12, 0.45)$, $\sigma_{A,\text{high}} \sim U(0.55, 1)$, $\sigma_V \sim U(0.3, 1)$, p_c and $p_{\text{switch}} \sim U(0.1, 0.9)$.
- **Bias, in both intervals:** $\ln \hat{v} = \ln v - c_{\text{noise}} - d \ln(v/0.5)$, with $c_{\text{low}} \sim U(0, 0.5)$, $c_{\text{high}} \sim U(-0.1, 0.4)$ and $d \sim U(-0.3, 0.6)$. These ranges come from the cross-modal blocks of P101, P102 and P104: c 0.06–0.50, d −0.23 to 0.60.
- **Main experiment as it now runs:** 9 conflicts (± 450 ms) × 2 noise levels, a 2D1U and a 1D2U staircase per condition, easy trials, whole frames at 120 Hz. 70 trials per condition, 1260 in all. (106 trials per condition, run only with σ_A free, gave the same results.)
- **Fits:** each model fitted three ways by BADS: no bias; d only; c_{low}, c_{high} and d. The biased fits also start from the no-bias solution, so they never fit worse.
- **What the unimodal blocks contribute** (the three scenarios):
  - *σ_A free:* σ_A is fitted from the bimodal data alone. $\sigma_V/(1-d)$, the visual σ the unimodal block measures, is fixed at its true value $\times\, e^{N(0,0.2)}$.
  - *σ_A known, log error 0.15 or 0.05:* each σ_A is estimated as the true value $\times\, e^{N(0,\text{err})}$ and added to the likelihood as a Gaussian penalty on $\ln \sigma_A$, which approximates the unimodal block's likelihood in a joint fit. σ_V is fixed with the same log error.

  All three scenarios use the same simulated data. For comparison, the QUEST+ simulations (`analysis/quest_simulation_results`) put the log error of a 100 + 20-trial block at about 0.23 and of the 150 + 20 used from 29 Oct at about 0.20, so 0.15 is optimistic (it would take about 300 trials per condition).

2 Results

2.1 c is not identified

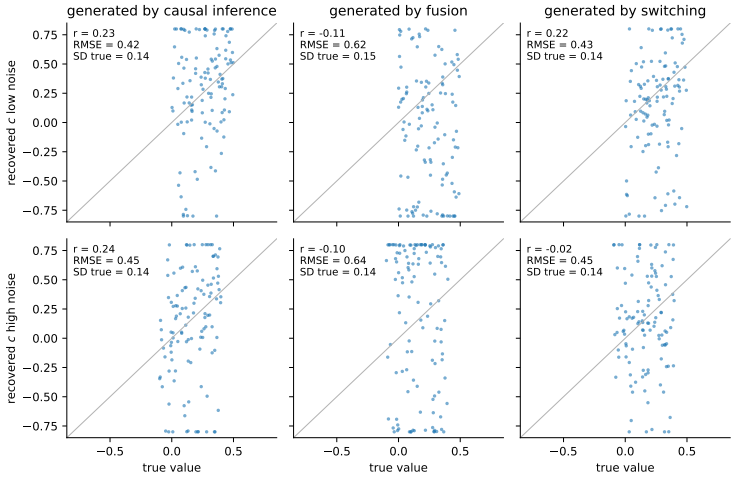


Figure 1: Recovered vs true c, from the fit with c and d free, by the generating model; σ_A known (log error 0.15). The other scenarios look the same: r between -0.11 and 0.38, RMSE 0.40–0.67 against a true SD of 0.14. Values beyond ± 0.8 are at the bounds.

2.2 d: fusion observers only, and only with precise σs

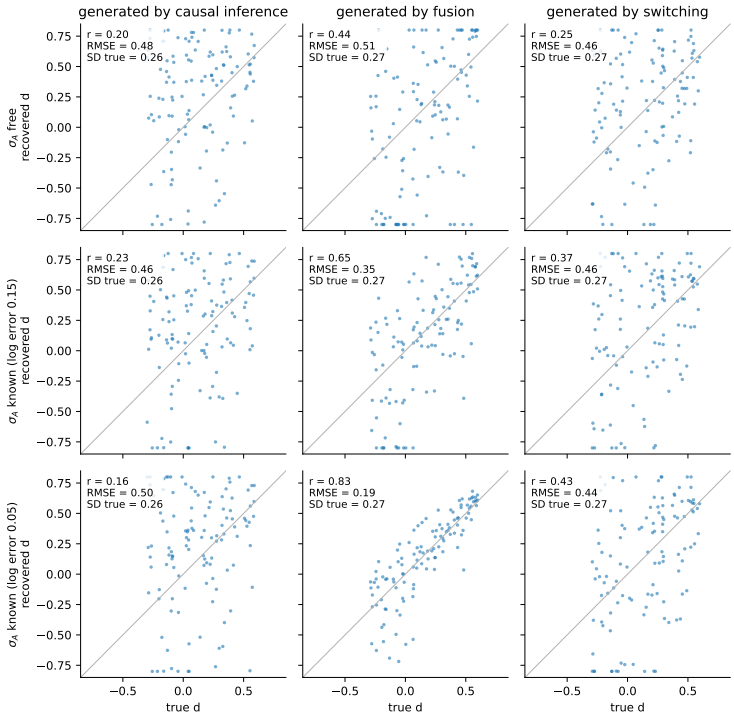


Figure 2: Recovered vs true d (fit with c and d free). Rows: what is known about σ_A. Only fusion observers with precisely known σs give a usable d. The d-only fit gives the same picture (fusion $r = 0.44$ / 0.65 / 0.83 across the three scenarios) and underestimates d for CI observers by 0.21–0.26 on average.

The reason is that, once σ_A and σ_V are known, the fused PSE shift is set by $w_V(1-d)$ times the conflict, so d follows from the size of the shift. In CI and switching, the size of the shift also depends on p_c or p_{switch}, so a smaller d and a smaller p_c look alike. This is also why AIC picks the d fit

for 98% of fusion observers with $|d| > 0.3$ when σ_A is known precisely. A large d is detectable in a fusion observer, but it cannot be measured in the others.

2.3 Model recovery

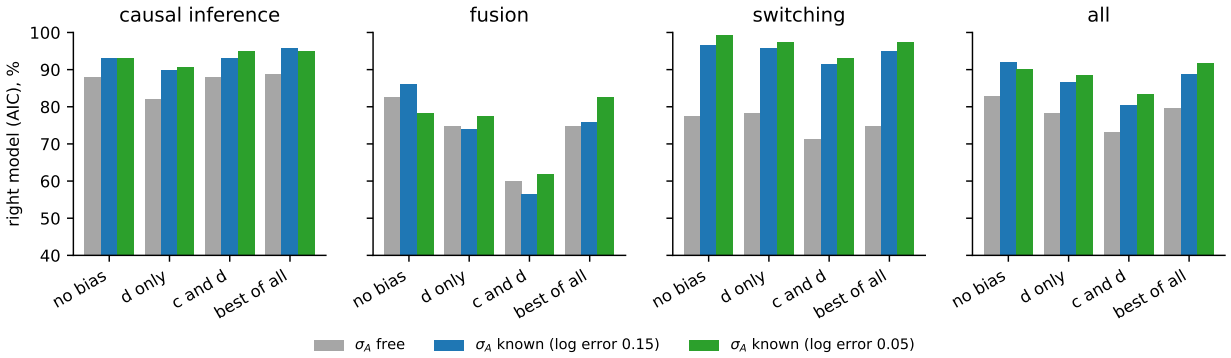


Figure 3: Right model by AIC. "Best of all" picks among all nine fits (3 models × 3 bias variants). Chance is 33%.

fit	scenario	CI	fusion	switching	all
no bias	σ_A free	88	83	77	83
	σ_A known, 0.15	93	86	97	**92**
	σ_A known, 0.05	93	78	99	90
d only	σ_A free	82	75	78	78
	σ_A known, 0.15	90	74	96	86
	σ_A known, 0.05	91	77	97	88
c and d	σ_A free	88	60	71	73
	σ_A known, 0.15	93	57	91	80
	σ_A known, 0.05	95	62	93	83
best of all	σ_A free	89	75	75	79
	σ_A known, 0.15	96	76	95	89
	σ_A known, 0.05	95	83	97	92

Table 1: Right model by AIC, % (115–116 participants per model; the SE of a single cell is about 3–4 points).

- **With σ_A free,** fusion and switching are mistaken for CI: 17% and 16% of the time (no-bias fits).
- **With σ_A known,** switching is almost always found (97%). The remaining error is fusion read as CI (10%).
- **Bias terms hurt fusion most.** With c and d free, CI absorbs fusion observers, and fusion recovery drops to 57–62%.
- **Very precise σs expose the missing d.** With log error 0.05 and no bias terms, recovery falls from 98% for $0 < d < 0.3$ to 80% for $d > 0.3$, and fusion overall falls to 78%. When σ_A is known precisely, a large unmodelled d no longer fits. At log error 0.15 the drop is smaller (96% to 89%).

2.4 p_c

	no bias		d only		c and d	
scenario	r	RMSE	r	RMSE	r	RMSE
σ_A free	0.81	0.167	0.78	0.171	0.63	0.222
σ_A known, 0.15	0.83	0.155	0.82	0.158	0.73	0.193
σ_A known, 0.05	0.88	0.127	0.83	0.156	0.78	0.176

Table 2: p_c recovered by the CI fit for CI participants (true SD 0.25). Mean p_c error of the no-bias fit by true d (σ_A known, 0.15): -0.12 for $d < 0$, -0.02 for $0 \le d \le 0.3$, $+0.02$ for $d > 0.3$. The d fit halves the first (-0.06) without lowering the RMSE.

3 Noise calibration and presentation order

`analysis/noise_calibration_order_check.py`. P105's calibration chose noise 1.67, where the unimodal auditory block then measured σ 1.6–1.9 instead of the target 0.75. P103 showed the same pattern: calibrated at 1.78, unimodal σ 2.3.

What happens. With the test second, discrimination stays sharp in loud noise. With the test first it falls to chance, with a bias towards calling the second interval longer.

- **P105 at 1.67, test second:** "test longer" for 4% of 100 ms tests and 94% of 950 ms tests.
- **P105 at 1.67, test first:** 24% and 38%.

The same pattern appears in every participant except P102 (Table 3). It is not a stimulus artefact: PsychoPy's Hamming option only ramps 5 ms at each end. It matches the known advantage of the standard coming first in duration discrimination (internal-reference account).

	calibrated	calibration, % correct		unimodal block at that level, σ		
	level	test first	test second	pooled	test first	test second
P101	0.53	58	88	0.42	0.86	0.17
P102	0.93	63	77	–	–	–
P103	1.78	68	82	2.31	3.20	0.48
P104	1.10	59	88	0.47	1.43	0.20
P105	1.67	55	92	1.58	7.8	0.03

Table 3: Calibration: % correct at ± 0.41 (332/752 ms) within a factor of 1.4 of the chosen level. Unimodal block: lognormal fits per order (lapse ≤ 0.1). Very large or very small σs are poorly determined, because QUEST+ placed those tests at the extremes. P102's unimodal block at the calibrated level was stopped after 8 trials. In a joint fit of all of P102's data, P102 is the one symmetric listener (σ 0.60 vs 0.56 at noise 0.5).

Why the calibration overshot. The calibration scores % correct at a fixed ± 0.41, pooled over orders. Test-first trials near chance plus test-second trials at about 95% already give the 70.7% target. That pooled value barely changes between noise 1 and 2 (P105: about 77% to 74%), so the staircases cannot find the level and drift upwards. The unimodal block instead fits one σ over both orders and tests of 80–950 ms. There, the test-first trials stay near chance even at a 9.5-fold difference, which needs a very large σ.

Starting point, trials and a cap. Each participant's data were fitted with a σ(noise) and a PSE per order, and the calibration was rerun 150 times on that fitted listener:

calibration	P101	P102	P103	P104	P105
before 28 Oct (start 0.3/1.6, 100 per staircase, cap 2.0)	0.70	1.07	1.24	1.32	1.44 (0.94–1.79)
both start at 0.5	0.61	0.93	1.06	1.20	1.36 (0.79–1.71)
200 trials per staircase	0.63	1.06	1.15	1.42	1.47 (1.11–1.85)
cap 1.5	0.64	0.91	1.03	0.98	1.17 (0.76–1.41)
cap 1.2, both start at 0.5 (since 28 Oct)	0.64	0.84	0.95	0.89	0.99 (0.75–1.11)

Table 4: Median level reached (80% range for P105). More trials do not help, the start point helps a little, and the cap does most of the work. The calibration summary now also reports % correct by order near the chosen level, and warns if test-first is below 60% and more than 25 points behind test-second.

Consequence for the models. With the test second, noise barely changes discrimination: P105's σ goes from 0.12 at noise 0.5 to 0.22 at 2.0, P103's from 0.28 to 0.41. The noise manipulation of auditory reliability therefore works mostly through the first interval. The bimodal models (and the simulations above) assume a single σ_A per noise level. If the bimodal data show the same order dependence, sharing a pooled unimodal σ_A with them may mislead, and the σ_A gain in Table 1 would not carry over.

4 Limitations

- The simulated observers have no order effect, and the σ_A penalty assumes the unimodal block measures the σ_A that the bimodal task uses (Section 3).
- σ_V's error is 0.2 in the σ_A-free scenario and 0.15 in the σ_A-known one, so part of the gain may come from σ_V. In the 28 Oct report, σ_V alone (free vs fixed) moved recovery by 11 points, while 0.2 vs 0.15 is a much smaller change.
- The bias ranges rest on three participants' cross-modal blocks.
- Fits are BADS local optima, with 3 starts plus the no-bias start. The bound hits for c show a flat likelihood, not a failed optimiser: the same fits recover the model parameters well.
- Only 70 trials per condition were simulated in the σ_A-known scenarios. With σ_A free, 106 trials gave the same bias recovery as 70.